

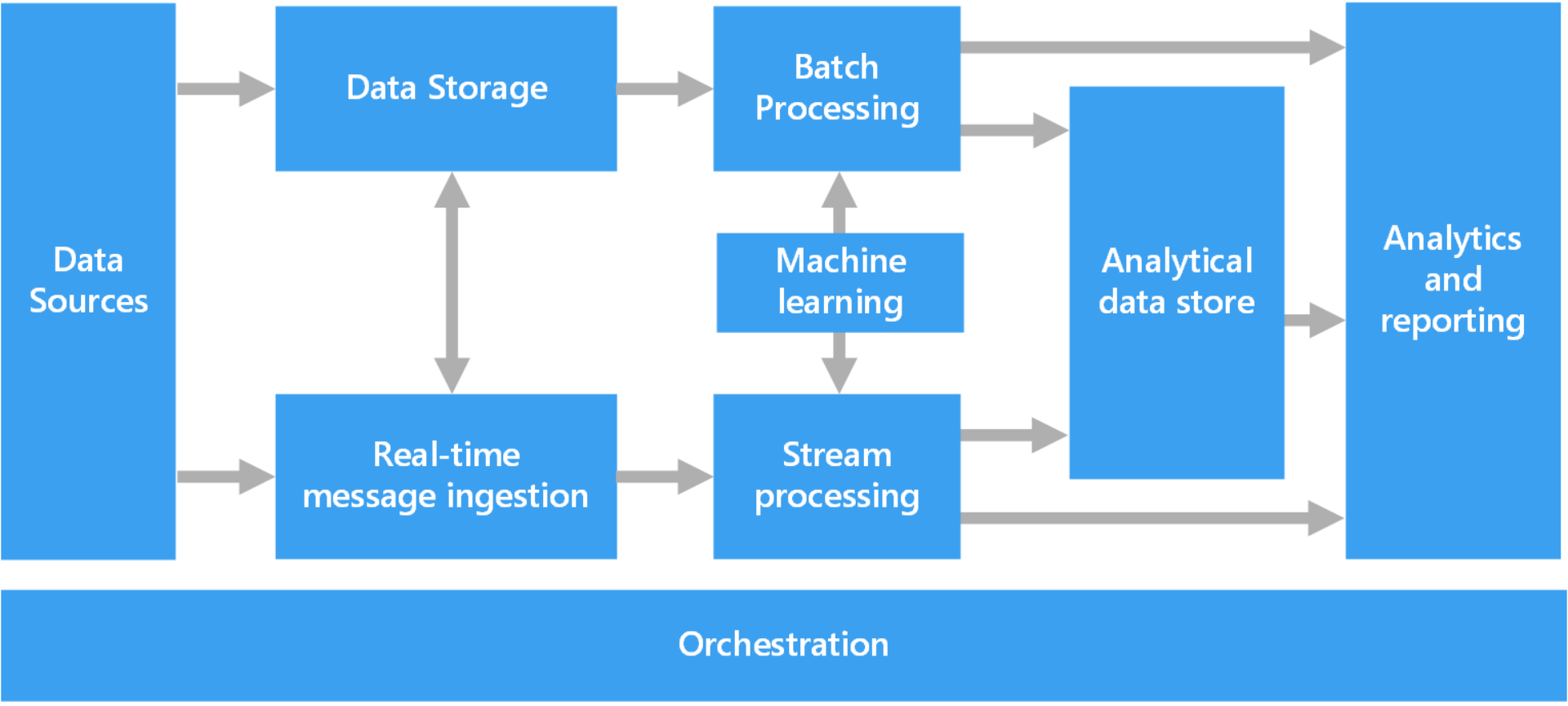


Proveedores de soluciones: Azure

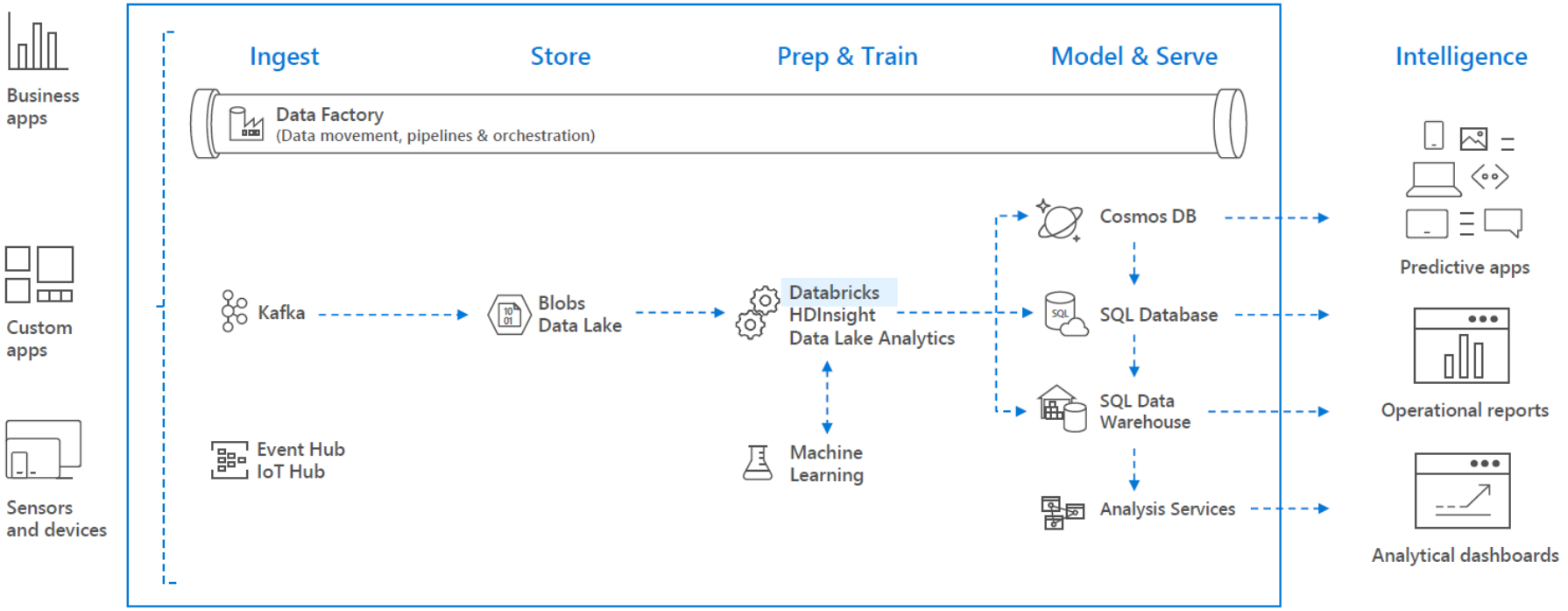
Rafael Pastor Vargas
Andrés Duque Fernández
SCC-ETSI Informática-UNED

https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/big-data

AZURE BIG DATA ARCHITECTURES



BIG DATA Y ANALITICA EN MICROSOFT AZURE



https://docs.microsoft.com/en-us/azure/#pivot=products&panel=analytics

MICROSOFT AZURE ANALYTICS (1/2)



SQL Data Warehouse

Elastic data warehouse as a service with enterprise-class features



Azure Databricks

Create an Azure Databricks workspace with Spark clusters



HDInsight

Provision cloud Hadoop, Spark, R Server, HBase, and Storm clusters



Data Factory

Orchestrate and manage data transformation and movement



Stream Analytics

Real-time data stream processing from millions of IoT devices



Data Lake Analytics

Distributed analytics service that makes big data easy



Event Hubs

Receive telemetry from millions of devices



Power BI Embedded

Embed fully interactive, stunning data visualizations in your applications



Azure Analysis Services

Enterprise grade analytics engine as a service

https://docs.microsoft.com/en-us/azure/#pivot=products&panel=analytics

MICROSOFT AZURE ANALYTICS (2/2)



Apache Spark for Azure HDInsight

Apache Spark in the cloud for mission critical deployments



Apache Storm for Azure HDInsight

Real-time stream processing made easy for big data



R Server for Azure HDInsight

Predictive analytics, machine learning, and statistical modeling for big data



Log Analytics

Collect, search, and visualize machine data from on-premises and cloud



Data Catalog

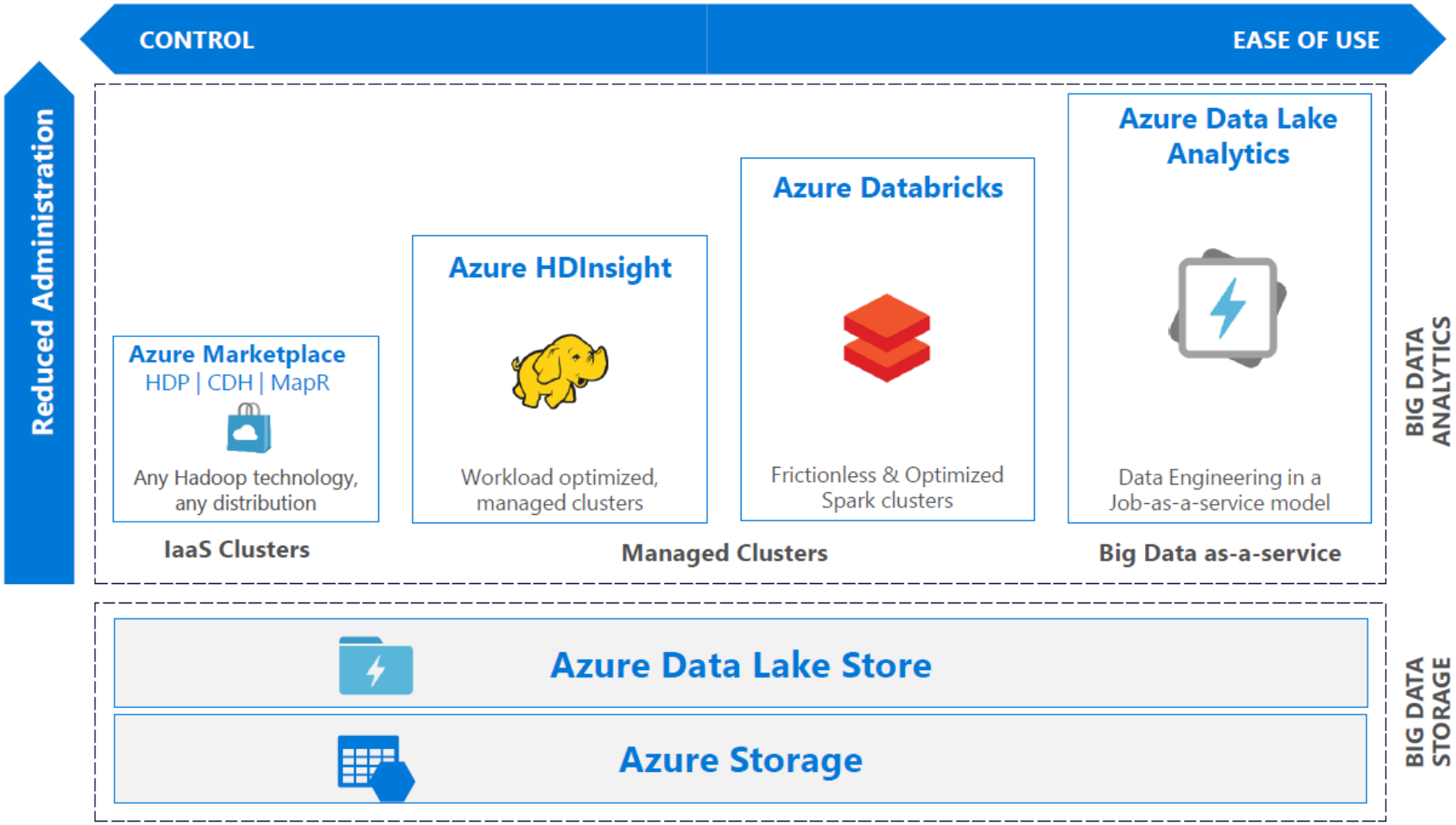
Get more value from your enterprise data assets



Data Lake Store

Hyperscale repository for big data analytics workloads

SOLUCIONES BIG DATA



https://docs.microsoft.com/en-us/azure/hdinsight/

HDINSIGHT



Managed Open Source Analytics for the cloud with a 99.9% SLA.

100% Open Source

Clusters up and running in minutes

63% lower TCO than deploy your own Hadoop on-premises

Separation of compute and store allows you to scale clusters to exponentially reduce costs

https://docs.microsoft.com/es-es/azure/hdinsight/hdinsight-component-versioning

HDINSIGHT, BASADO EN HORTONWORKS

Component	HDInsight 4.0	HDInsight 3.6 (Default)	HDInsight 3.5	HDInsight 3.4	HDInsight 3.3	HDInsight 3.2
Apache Hadoop and YARN	3.1.1	2.7.3	2.7.3	2.7.1	2.7.1	2.6.0
Apache Tez	0.9.1	0.7.0	0.7.0	0.7.0	0.7.0	0.5.2
Apache Pig	0.16.0	0.16.0	0.16.0	0.15.0	0.15.0	0.14.0
Apache Hive	3.1.0	2.1.0, 1.2.1	-	-	-	-
Apache Tez Hive2	-	0.8.4	-	-	-	-
Apache Ranger	1.1.0	0.7.0	0.6.0	-	-	-
Apache HBase	2.0.1	1.1.2	1.1.2	1.1.2	1.1.1	0.98.4
Apache Sqoop	1.4.7	1.4.6	1.4.6	1.4.6	1.4.6	1.4.5
Apache Oozie	4.3.1	4.2.0	4.2.0	4.2.0	4.2.0	4.1.0
Apache Zookeeper	3.4.6	3.4.6	3.4.6	3.4.6	3.4.6	3.4.6
Apache Storm	-	1.1.0	1.0.1	0.10.0	0.10.0	0.9.3
Apache Mahout	-	0.9.0+	0.9.0+	0.9.0+	0.9.0+	0.9.0
Apache Phoenix	5	4.7.0	4.7.0	4.4.0	4.4.0	4.2.0
Apache Spark	2.3.1, 2.4	2.3.0, 2.2.0, 2.1.0	1.6.2, 2.0	1.6.0	1.5.2	1.3.1 (Windows only)
Apache Livy	0.5	0.4, 0.4, 0.3	0.3	0.3	0.2	-
Apache Kafka	1.1.1, 2.1	1.1, 1.0 * (See Note below)	0.10.0	0.9.0	-	-
Apache Ambari	2.7.0	2.6.0	2.4.0	2.2.1	2.1.0	-
Apache Zeppelin	0.8.0	0.7.0	-	-	-	-
Mono	4.2.1	4.2.1	4.2.1	3.2.8	-	-

INTEGRACIÓN CON TERCEROS



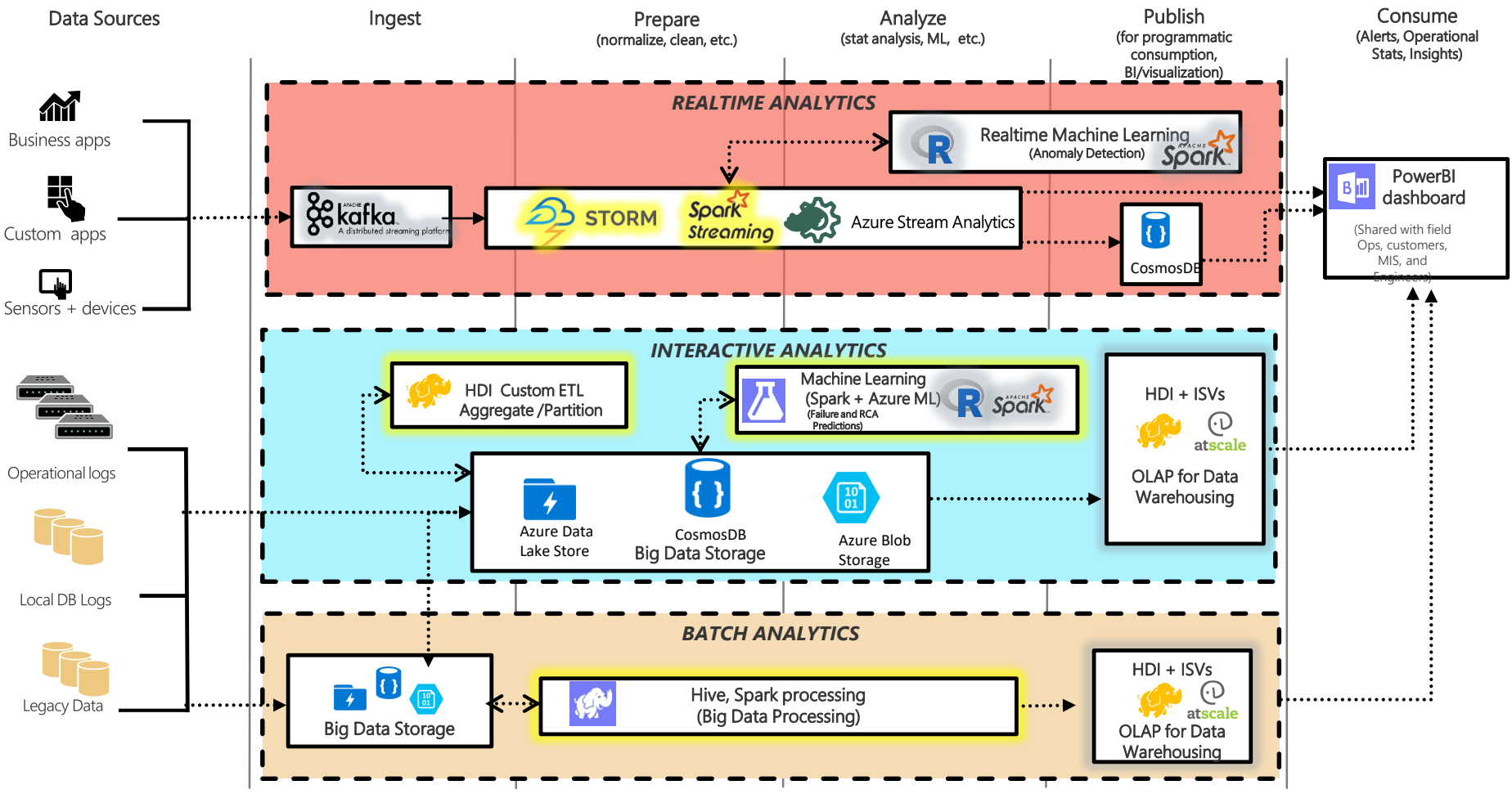
Easy ISV integration as you deploy the cluster

Datameer atscale CASK StreamSets™

https://docs.microsoft.com/es-es/azure/hdinsight/hadoop/apache-hadoop-introduction

ESCENARIOS DE USO (1/2)

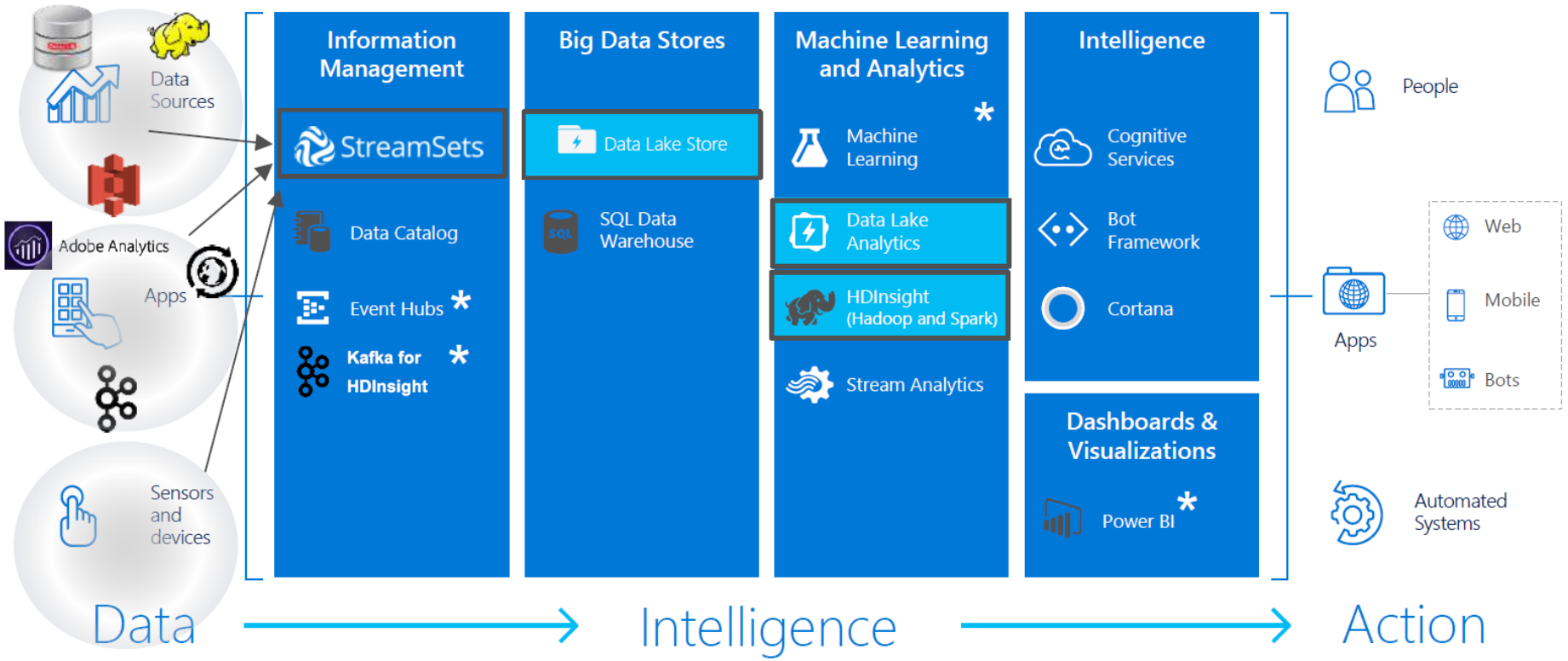
Reference Big Data Analytics Pipeline



ESCENARIOS DE USO (2/2)

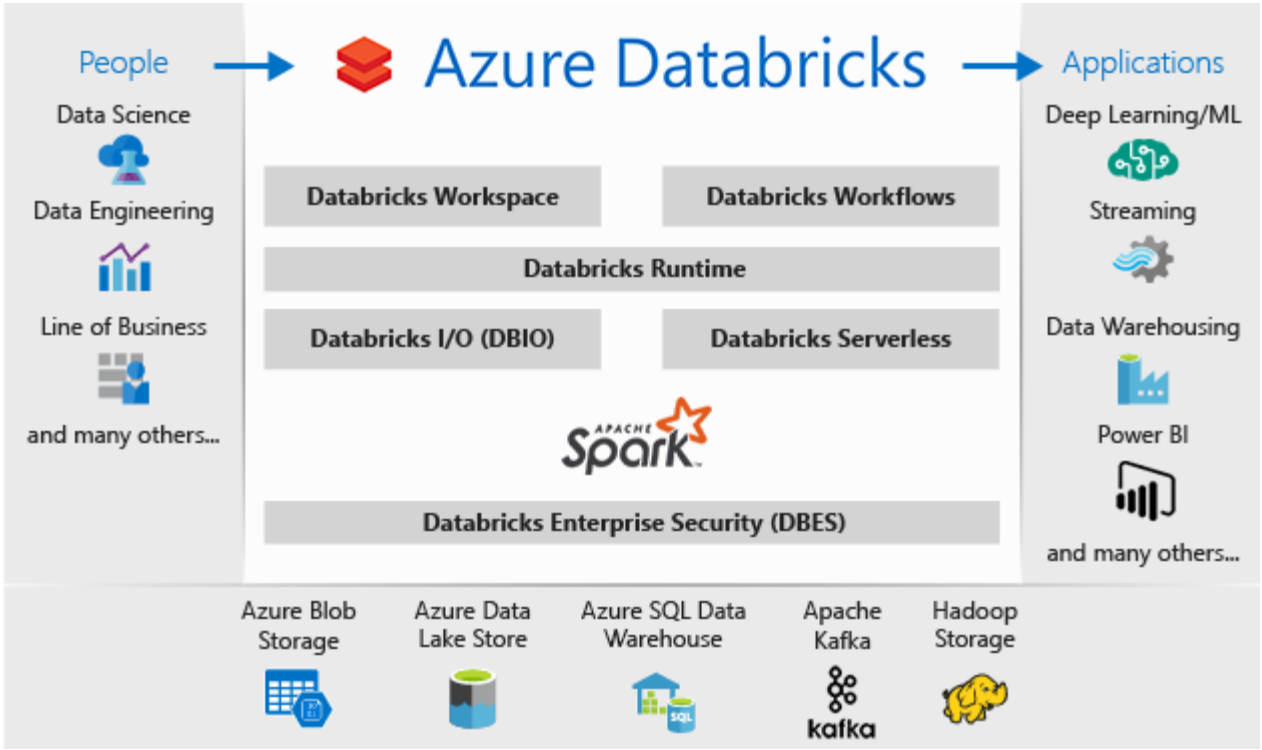
STREAMSETS DATA COLLECTOR

Deep Integration with Microsoft Azure



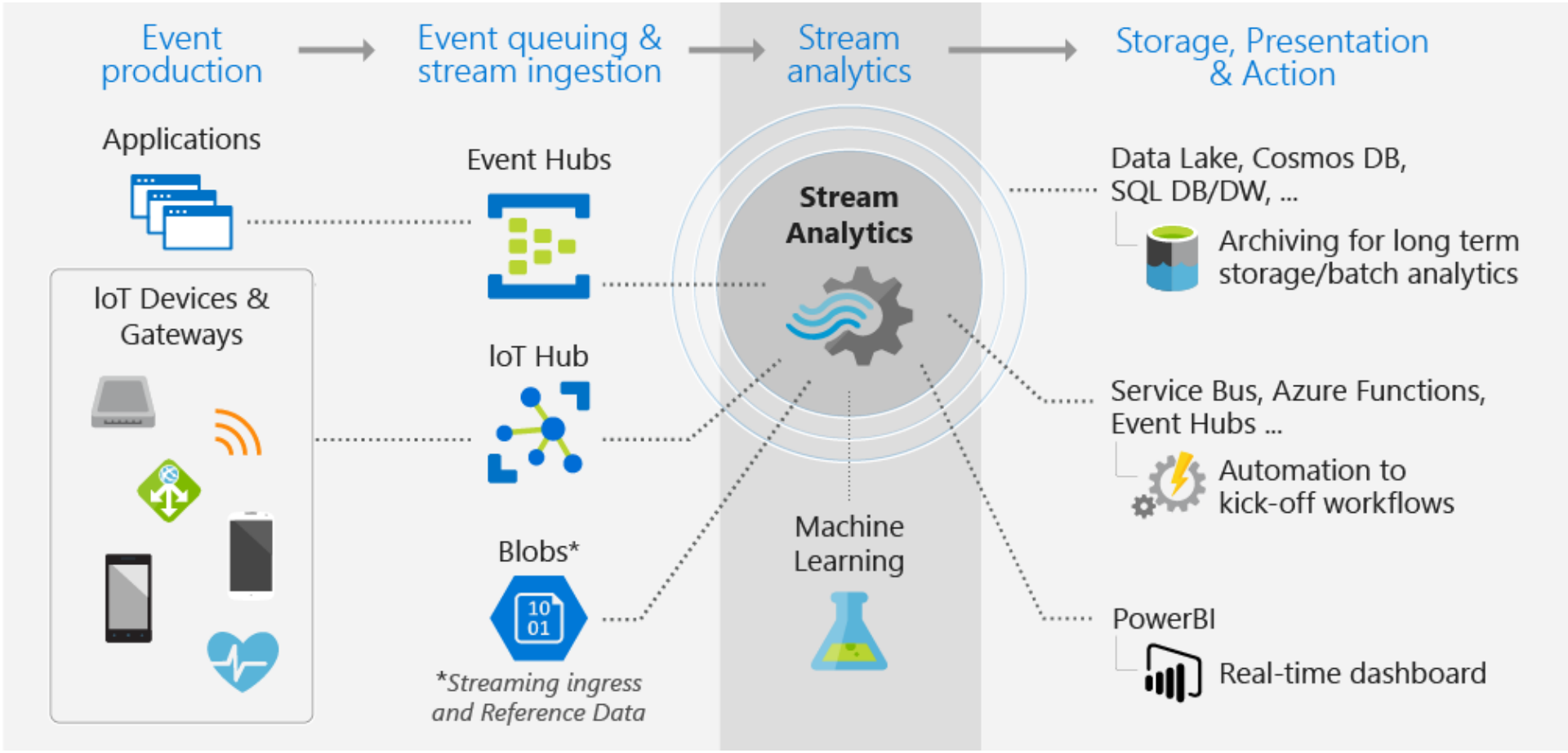
DATABRICKS

- https://docs.microsoft.com/en-us/azure/azure-databricks/quickstart-create-databricks-workspace-portal



https://docs.microsoft.com/en-us/azure/stream-analytics/stream-analytics-introduction

AZURE STREAMING ANALYTICS



- https://docs.microsoft.com/en-us/learn/paths/implement-data-streaming-with-asa/

MACHINE LEARNING AND AI

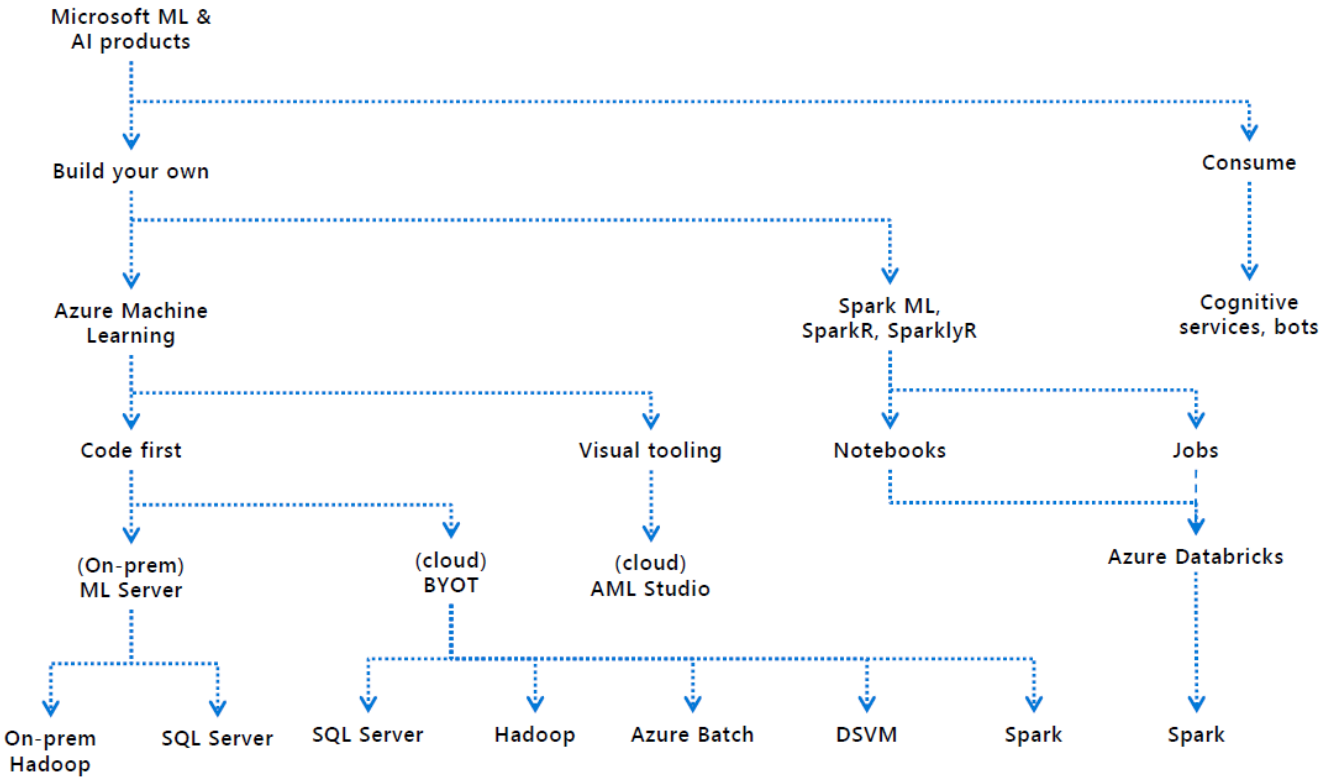
When to use what

Build your own or consume pre-trained models?

Which experience do you want?

Deployment target

What engines do you want to use?



https://docs.microsoft.com/en-us/azure/#pivot=products&panel=ai

MICROSOFT AZURE AI



Azure Databricks

Create an Azure Databricks workspace with Spark clusters



Azure Batch AI

Easily experiment and train your deep learning and AI models in parallel at scale



Azure Bot Service

Intelligent, serverless bot service that scales on demand



Microsoft Genomics

Power genome sequencing and research insights



Machine Learning Services

Build, deploy, and manage machine learning and AI models



Machine Learning Studio

Easily build, deploy, and manage predictive analytics solutions



Cognitive Services

Infuse apps, websites, and bots with intelligence to see, hear, speak, and understand



Azure Search

Fully-managed search-as-a-service

Cognitive Services APIs

Vision

- Computer Vision API
- Content Moderator
- Custom Vision Service PREVIEW
- Emotion API PREVIEW
- Face API
- Video Indexer PREVIEW

Speech

- Bing Speech API
- Custom Speech Service PREVIEW
- Speaker Recognition API PREVIEW
- Translator Speech API

Language

- Bing Spell Check API
- Language Understanding (LUIS)
- Linguistic Analysis API PREVIEW
- Text Analytics API
- Translator Text API
- Web Language Model API PREVIEW

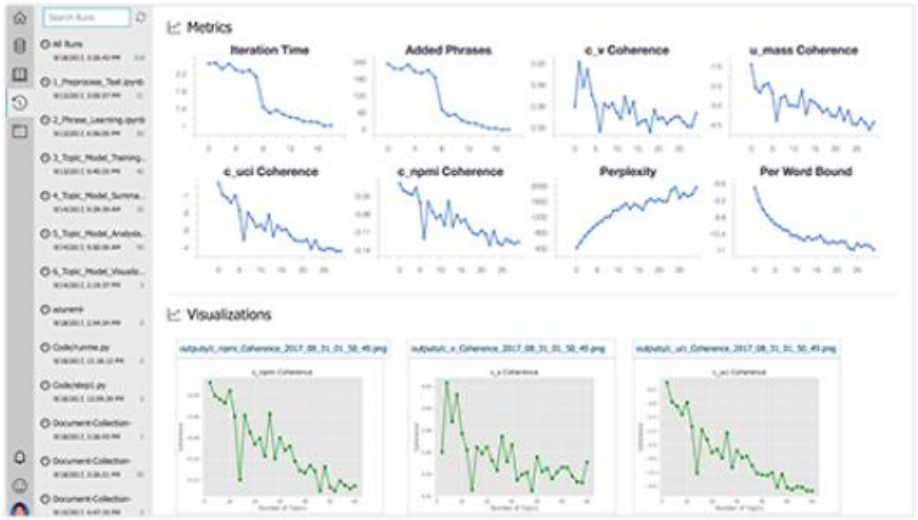
Knowledge

- Custom Decision Service PREVIEW
- QnA Maker API PREVIEW
- Recommendations API PREVIEW
- Knowledge Exploration Service (KES)
- Academic Knowledge API
- Entity Linking Intelligence Service API
- URL Preview
- Answer Search
- Anomaly Finder
- Conversation Learner

Search

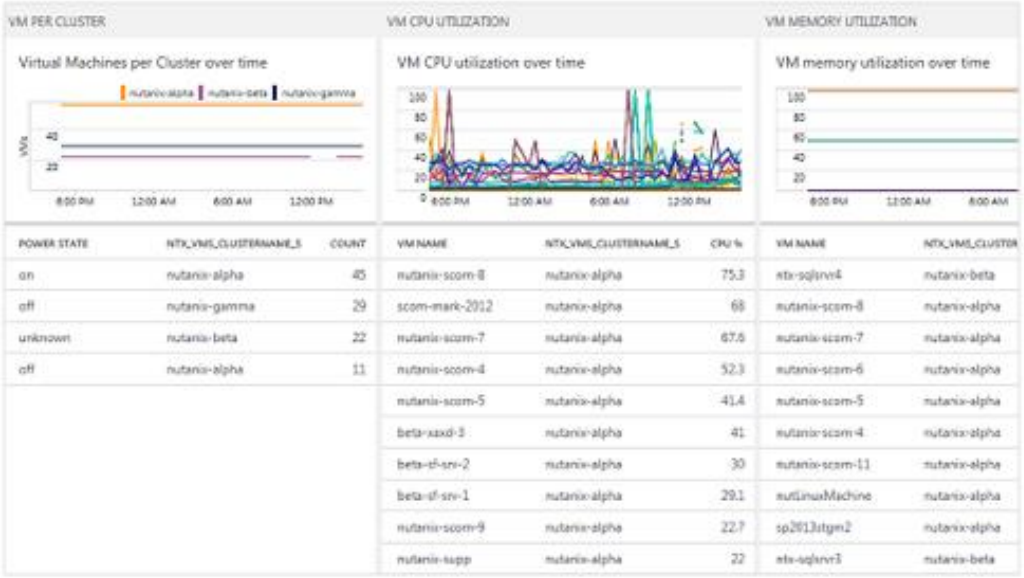
- Bing News Search API
- Bing Video Search API
- Bing Web Search API
- Bing Visual Search API
- Bing Autosuggest API
- Bing Custom Search API
- Bing Entity Search API
- Bing Image Search API

MICROSOFT MACHINE LEARNING



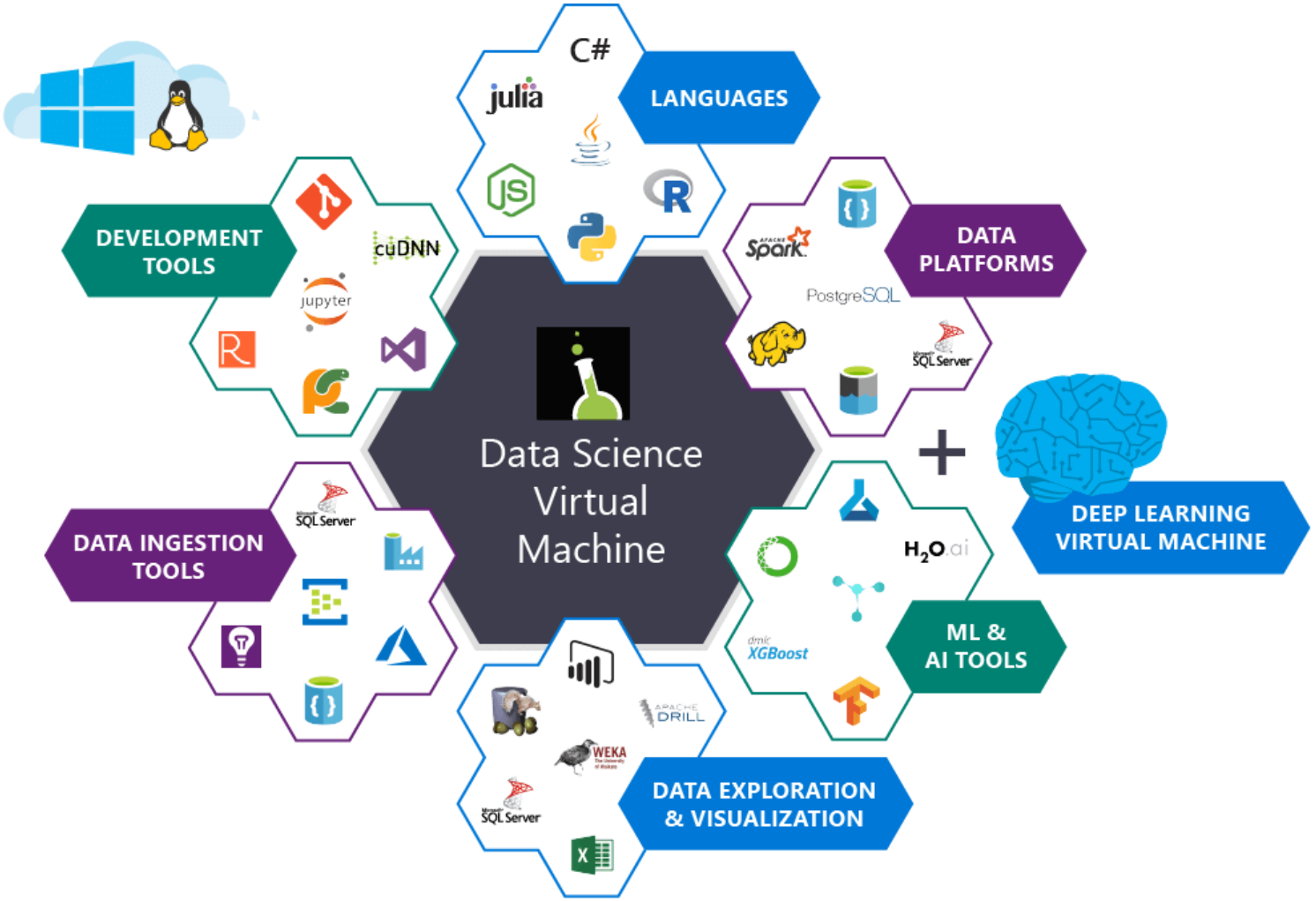
- Machine Learning a escala de macrodatos
- Implementación de inteligencia artificial basada en contenedores desde la nube hasta el perímetro
- Rápida, escalado horizontal, experimentación en colaboración
- Doma de datos (wrangling) con inteligencia artificial
- Spark, Docker, Cognitive Toolkit, TensorFlow, Caffe, etc.

MÁQUINAS VIRTUALES PRECONFIGURADAS

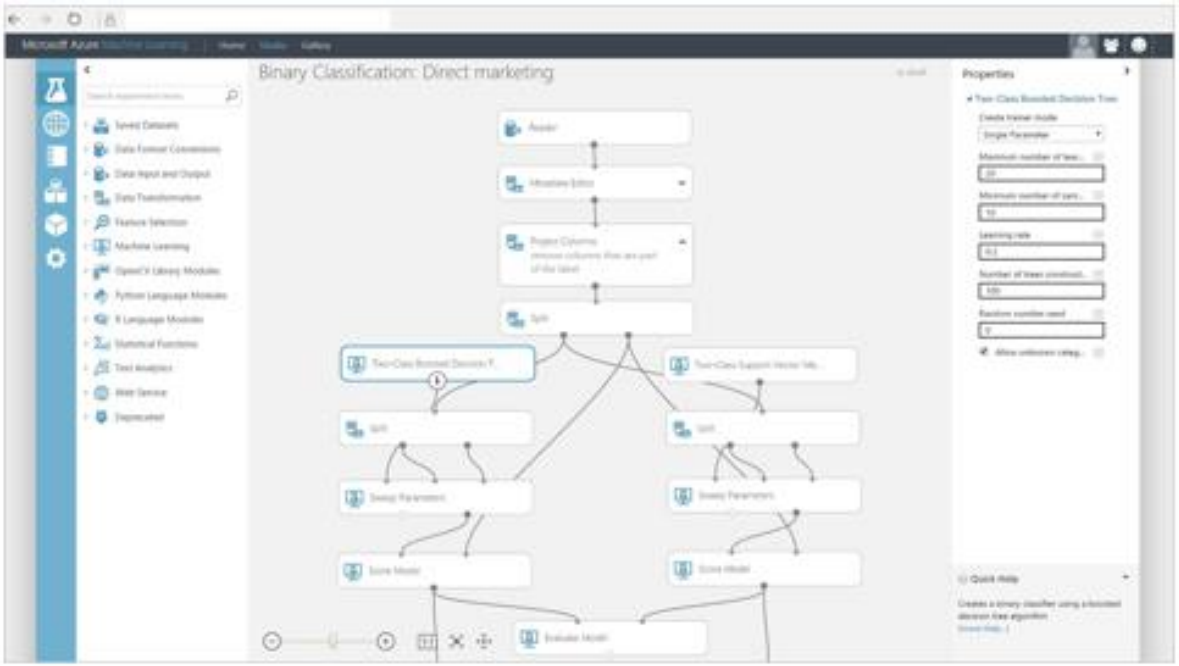


- ✓ Entornos preconfigurados para científicos de datos
- ✓ Perfecta integración con los servicios de Azure
- ✓ Cualquier tamaño de máquina virtual, escalado horizontal con Azure Batch, gran aumento del proceso
- ✓ Compatibilidad con GPU y los marcos DNN más populares
- ✓ Extensible por el usuario

MV: SOFTWARE


LANGUAGES
C#
julia
DEVELOPMENT TOOLS
cuDNN
jupyter
DATA PLATFORMS
Spark
PostgreSQL
SQL Server
Data Science Virtual Machine
DEEP LEARNING VIRTUAL MACHINE
DATA INGESTION TOOLS
SQL Server
ML & AI TOOLS
H2O.ai
XGBoost
DATA EXPLORATION & VISUALIZATION
APACHE DRILL
WEKA
SQL Server

AZURE MACHINE LEARNING STUDIO



- ✔ Desarrollo con funcionalidad de “arrastrar y colocar”, sin servidor
- ✔ Experimentación intuitiva sin código
- ✔ Implementación de servicios web en minutos

Azure Machine Learning Studio Capabilities Overview

https://studio.azureml.net

Guest Access Workspace: Free trial access without logging in.
Free Workspace: Free persisted access, no Azure subscription needed.
Standard Workspace: Full access with SLA under an Azure subscription.

Machine Learning in ML Studio

Anomaly Detection
- One-class Support Vector Machine
- Principal Component Analysis-based Anomaly Detection
- Time Series Anomaly Detection*

Classification
- Two-class Classification
  - Averaged Perceptron
  - Bayes Point Machine
  - Boosted Decision Tree
  - Decision Forest
  - Decision Jungle
  - Logistic Regression
  - Neural Network
  - Support Vector Machine
- Multi-class Classification
  - Decision Forest
  - Decision Jungle
  - Logistic Regression
  - Neural Network
  - One-vs-all

Clustering
- K-means Clustering

Recommendation
- Matchbox Recommender

Regression
- Bayesian Linear Regression
- Boosted Decision Tree
- Decision Forest
- Fast Forest Quantile Regression
- Linear Regression
- Neural Network Regression
- Ordinal Regression
- Poisson Regression

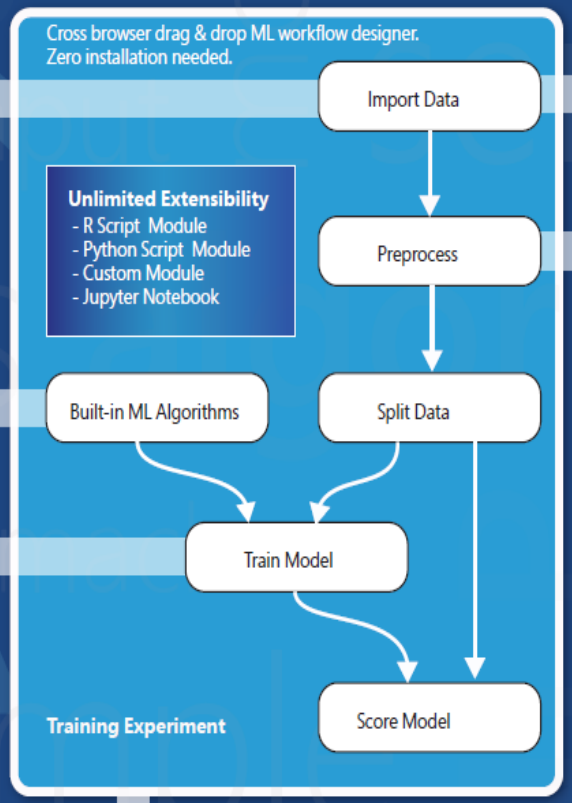
Statistical Functions
- Descriptive Statistics
- Hypothesis Testing T-Test
- Linear Correlation
- Probability Function Evaluation

Text Analytics
- Feature Hashing
- Named Entity Recognition
- Vowpal Wabbit

Computer Vision
- OpenCV Library

Data/Model Visualization
- Scatterplots
- Bar Charts
- Box plots
- Histogram
- R and Python Plotting Libraries
- REPL with Jupyter Notebook
- ROC, Precision/Recall, Lift
- Confusion Matrix
- Decision Tree*

Training
- Cross Validation
- Retraining
- Parameter Sweep



Data Source
- Azure Blob Storage
- Azure SQL DB
- Azure SQL DW*
- Azure Table
- Desktop Direct Upload
- Hadoop Hive Query
- Manual Data Entry
- OData Feed
- On-prem SQL Server*
- Web URL (HTTP)

Data Format
- ARFF
- CSV
- SVMLight
- TSV
- Excel
- ZIP

Data Preparation
- Clean Missing Data
- Clip Outliers
- Edit Metadata
- Feature Selection
- Filter
- Learning with Counts
- Normalize Data
- Partition and Sample
- Principal Component Analysis
- Quantize Data
- SQLite Transformation
- Synthetic Minority Oversampling Technique

Enterprise Grade Cloud Service
- SLA: 99.95% Guaranteed Up-time
- Azure AD Authentication
- Compute at Large Scale
- Multi-geo Availability
- Regulatory Compliance*

One-click Operationalization

Predictive Experiment

Make Prediction with Elastic APIs
- Request-Response Service (RRS)
- Batch Execution Service (BES)
- Retraining API

Community
- Gallery (http://gallery.azureml.net)
- Samples & Templates
- Workspace Sharing and Collaboration
- Live Chat & MSDN Forum Support

* Feature Coming Soon

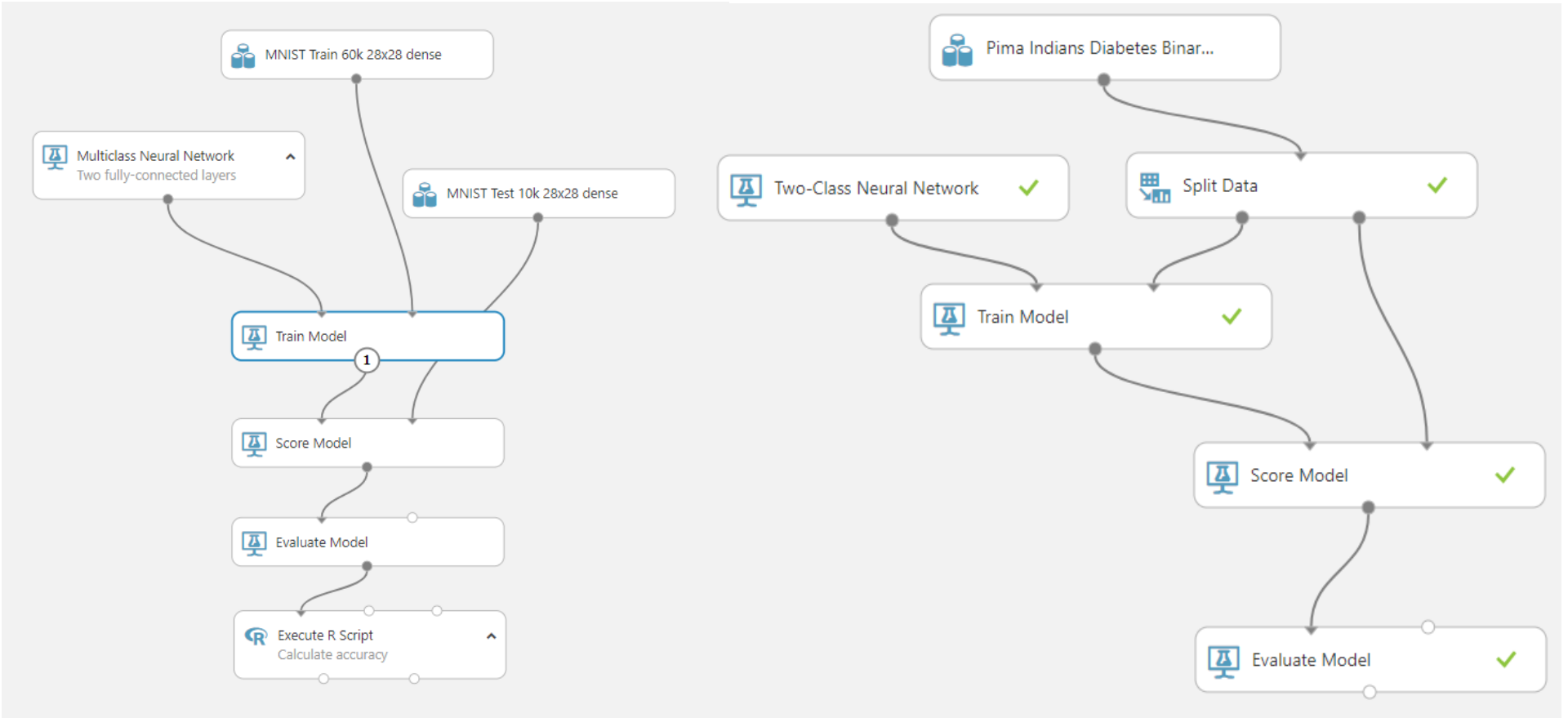
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Created by the Azure Machine Learning Team Email: AzurePoster@microsoft.com

Download this poster: http://aka.ms/MLStudioOverview

Microsoft

DEMO


MNIST Train 60k 28x28 dense
Multiclass Neural Network
Two fully-connected layers
MNIST Test 10k 28x28 dense
Train Model
1
Score Model
Evaluate Model
Execute R Script
Calculate accuracy
Pima Indians Diabetes Binar...
Two-Class Neural Network
Split Data
Train Model
Score Model
Evaluate Model

EJERCICIO

- https://docs.microsoft.com/es-es/learn/paths/publish-experiment-with-ml-studio/
- https://docs.microsoft.com/es-es/learn/modules/classify-user-feedback-with-the-text-analytics-api/index

RECURSOS EDUCACIÓN

- https://docs.microsoft.com/en-us/learn/azure/
- https://www.microsoft.com/handsonlabs
- https://portal.azure.com/#blade/Microsoft_Azure_Education/EducationMenuBlade/overview